CAD-CBA: A Class-Aware Distilled CNN–BiLSTM for Multi-Objective IoT Intrusion Detection with Operation-Matched CUDA Acceleration

Camera-ready **local-complete** draft · Method freeze CAD-CBA-v1 · Option A CUDA · Claims 59 · Champion md5 `80a90f7cc210276300eaa90173a5a385`
Authority: CLAIMS_REGISTRY + benchmarks/results · DICC multi-GPU cells TBD (never invent)

Abstract

(1) Background. IoT intrusion detection must handle extreme class imbalance and low-latency edge constraints. Tree ensembles often top pure accuracy on static tabular flows, while deep models are preferred for GPU deployment and systems co-design — yet unfair protocols and framework-only acceleration claims obscure real trade-offs.

(2) Gap. Prior work often siloes detection quality from deployment metrics, tunes without a sealed test, or reports single-shot laptop latencies as multi-platform facts. Full-pipeline “Custom CUDA vs full model” claims are invalid when kernels implement only operation-matched blocks (**Option A**).

(3) Method. We freeze **CAD-CBA-v1**: V3 CNN–BiLSTM–Attention; **ensemble** KD (mean RF+XGB+LGBM soft labels, α=**0.6**, T=**10**); **focal** loss ($\gamma\approx$**1.92**); Optuna train HPs (`config/hpo_best.yaml`); **shuffle** sampler; **argmax** decode. Evaluation uses protocol `botiot_v1` with fair classical/neural baselines, ablations, Pareto analysis, local multi-session systems ranges, scoped XAI, and a ToN recipe-transfer honesty check.

(4) Results. BoT sealed multi-seed TEST (n=5): macro-F1 **0.9780±0.0033**; min-cls **0.9292**; Theft **1.0000**. Protocol LGBM val **0.9818** remains pure-F1 ceiling; RF val **0.9778**; published RF **0.9864** is a different-pipeline dual bar. Ablation A7 **0.9699** tops package path; A4 attn+CE **0.7378** ■ A3 **0.9493**. Multi-obj composite #1 G6 **0.9056** @ **4.33** µs. Local systems (WP6b, RTX 3050, n=5): energy **0.920–0.943** mJ/flow; PT@256 **24.15–25.68** µs/sample; CUDA pipeline **565–570** µs (Option A); peak **322.2** MiB. XAI: dispatch p99 **16.60** µs; rank corr **0.9636**; faith **0.5109**; free-form feature-mention **0.333** → no full explainable claim. ToN 13-feat test **0.8110** vs RF **0.9393**.

(5) Contribution. Under sealed, protocol-fair evaluation, CAD-CBA-v1 delivers **near-RF multi-seed test detection** with **strong minority (Theft) recognition**, while the primary scientific claim is a **valid accuracy–efficiency multi-objective story** with **operation-matched CUDA** and **local multi-session ranges** — not pure-F1 supremacy over LightGBM and not multi-GPU portability until DICC completes.

Keywords: IoT intrusion detection; class imbalance; knowledge distillation; multi-objective evaluation; CUDA acceleration; BoT-IoT; protocol-fair baselines.

1. Introduction

Edge and campus IoT deployments need detectors that remain robust under severe class imbalance, support GPU inference with predictable latency and energy, and admit honest comparison against strong classical baselines on the **same** data protocol. Much of the literature optimises one axis and then generalises beyond the measured hardware and training recipe.

1.1 Research questions (locked answers)

RQ	Locked answer
RQ1 Detection	Test 0.9780±0.0033 ≈ RF val 0.9778; does not beat LGBM 0.9818 or published RF 0.9864
RQ2 Minority	Yes: Theft mean 1.0; min-cls mean 0.9292 (n=5 test)
RQ3 Latency/mem	Yes local: PT@256 24.15–25.68 µs; energy 0.920–0.943 mJ/flow; peak 322.2 MiB
RQ4 Across GPUs	Not yet claimable — DICC open
RQ5 XAI value	Dispatch + structured yes; free-form LLM no (J10 DROP_FULL)

1.2 Contributions

- CAD-CBA-v1 method package frozen and sealed multi-seed tested.
- Protocol-fair classical and neural baselines with dual-bar honesty vs published RF.
- Ablation + negative results (attention-alone hurts; C* probes rejected).
- Multi-objective Pareto and local multi-session systems ranges under Option A.
- Scoped XAI (dispatch + structured); full LLM-explainable title rejected.
- ToN recipe-transfer honesty on 13-feat data with RF gap disclosed.

1.3 Explicit non-claims

We do not claim pure-F1 supremacy over protocol LightGBM; full Custom-CUDA vs full V3 parity; multi-GPU/multi-day portability without DICC; or full free-form LLM explainability.

2. Related work and gaps

Deep hybrid CNN–RNN IDS models are common on IoT flow features but often omit sealed-test discipline and protocol-matched classical baselines. Imbalance methods (focal, CB losses, SMOTE, thresholds) are widespread; we compare them under one protocol. Tree-to-student distillation is attractive for deployment; we rank teachers by **student** quality. Systems papers risk invalid full-pipeline CUDA claims when kernels are operation-matched only. LLM-XAI work often conflates dispatch with generation and lacks faithfulness metrics.

Gap	Our response
Test used in selection	Sealed until B14 user lock
Classical on other pipelines	Protocol-fair same split + dual bar
Single-shot latency headline	Multi-session ranges n=5
Full-pipeline CUDA vs PT false parity	Option A + fidelity PASS
Novelty without ablation	A1–A7 + C* negatives
Full LLM-XAI in title	Metrics first; J10 drop full claim
Cross-GPU from one laptop	Explicit TBD until DICC

3. Method: CAD-CBA-v1

Input: 10 BoT-IoT flow features → linear projection 64 → reshape **[2, 32]** → CNN 64→128 (k=3, BN, ReLU, pool) → BiLSTM 128→64 → multi-head attention (4 heads) → dense 64 → 5-class softmax. Package path parameters: **530,181**.

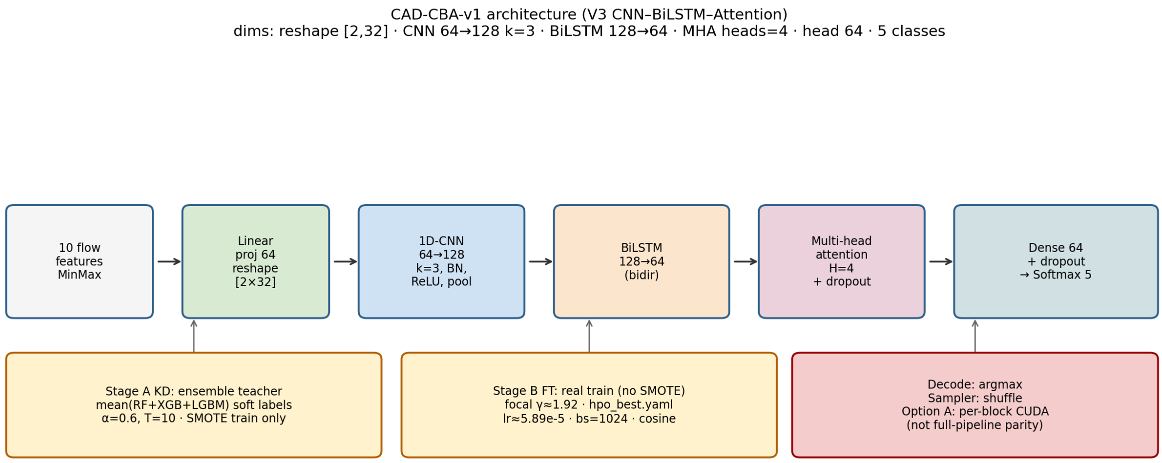

Figure 1. CAD-CBA-v1 architecture and training stages (config dims + hpo_best train HPs).

3.1 Frozen training recipe

Stage	Settings
A KD	Teacher = mean(RF+XGB+LGBM); α=0.6, T=10; SMOTE train only
B FT	No SMOTE; focal $\gamma\approx$1.9166; lr≈5.893e-5; bs=1024; cosine; shuffle
Decode	argmax (val threshold search: no lift)
B14 init	path A: distill a0.6_T10_focal2 + hpo_best FT; seeds 42–46
Champion	best_model_botiot_twostage.pth md5 80a90f7…a385 (systems/XAI)

Rejected for package: multi-scale CNN (0.9167), gated fusion (0.9132), SupCon (0.7732), ASL (0.8012), neural teacher student (0.8513), CB focal, stratified batching (0.9209■0.9791).

4. Experimental protocol

Protocol `botiot_v1`: official test sealed for selection. Primary metric macro-F1; always log min-cls, Theft, bal-acc, per-class F1. Seeds 42–46. Local GPU: RTX 3050 6GB; WP6b warm-up 50. Classical G1–G5 and neural G6–G12 on same splits.

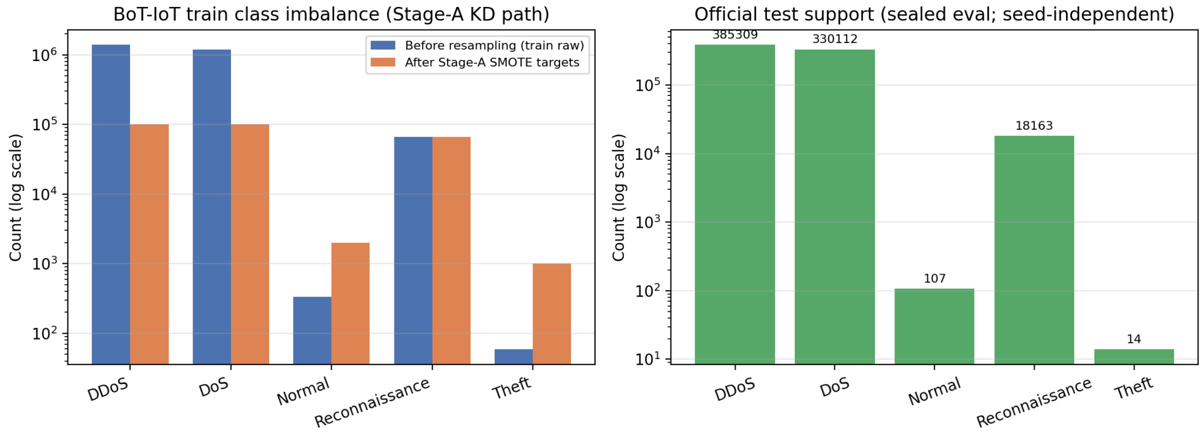

Figure 2. Train class imbalance (Stage-A SMOTE targets) and official test support (log scale). Theft test n=14.

5. Results

5.1 Sealed multi-seed BoT TEST (B14)

Seed	val F1	test F1	test min-cls	test Theft
42	0.9791	0.9787	0.9333	1.0000
43	0.9587	0.9798	0.9369	1.0000
44	0.9797	0.9798	0.9375	1.0000
45	0.9787	0.9722	0.9014	1.0000
46	0.9483	0.9796	0.9369	1.0000
Mean±std	0.9689±0.0145	0.9780±0.0033	0.9292	1.0000

Table 1. Source: sealed_test/summary.json. Seed46 val weaker (0.9483) but test strong (0.9796) — both reported. Champion md5 unchanged.

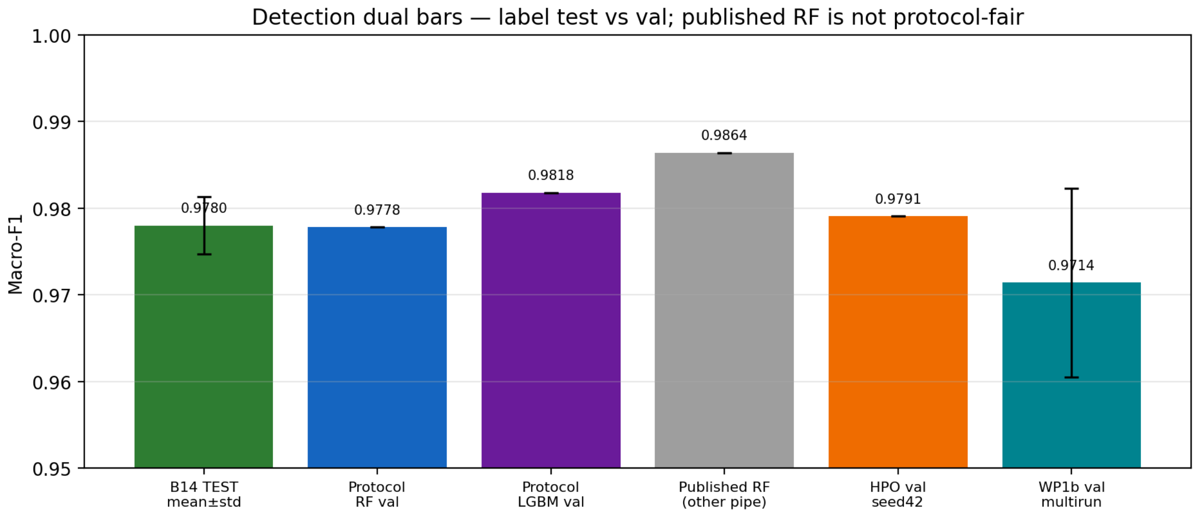

Figure 3. Dual accuracy bars: sealed test vs protocol classical val vs published RF (other pipeline).

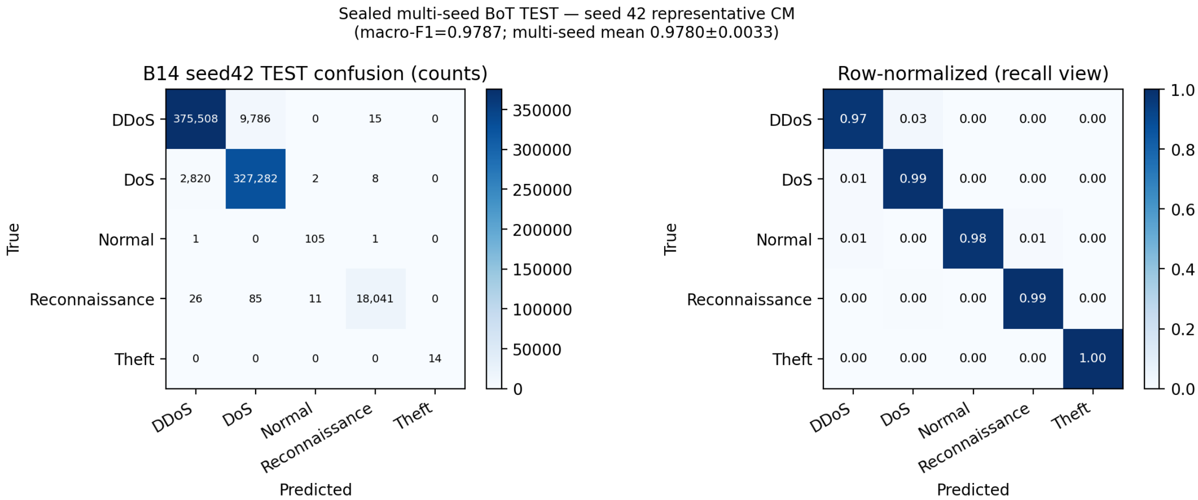

Figure 4. Representative sealed-test confusion matrix (seed 42; test macro-F1 0.9787).

5.2 Protocol-fair classical (val)

Model	val macro-F1	Notes
LGBM (balanced)	0.9818	Protocol pure-F1 ceiling
RF	0.9778	Protocol-fair
XGB	0.9762	
LR	0.5231	
LinearSVC	0.4268	Weak under imbalance
Published RF	0.9864	Different pipeline — dual bar only

Table 2. RQ1: sealed test near RF; LGBM still tops pure F1.

5.3 Neural baselines (val, equal budget)

Model	val macro-F1
G11 CNN–BiLSTM	0.9493
G6 MLP	0.9285
G10 CNN–LSTM	0.8159
G8 LSTM	0.8099
G9 BiLSTM	0.8058
G7 1D-CNN	0.6221
G12 Transformer	0.5808

Table 3. Transformer is not a free win under equal CE budget.

5.4 Ablation ladder

Row	val macro-F1	Role
A7 full CAD-CBA-v1	0.9699	Package top
A3 cnn_bilstm CE	0.9493	Strong backbone
A6 +ens KD	0.9346	KD lift
A5 attn+focal	0.8684	Focal helps vs A4
A2 bilstm	0.8058	
A4 attn+CE	0.7378	Attention not free
A1 cnn only	0.6221	Weak

Table 4. Novelty is package composition under protocol, not attention alone.

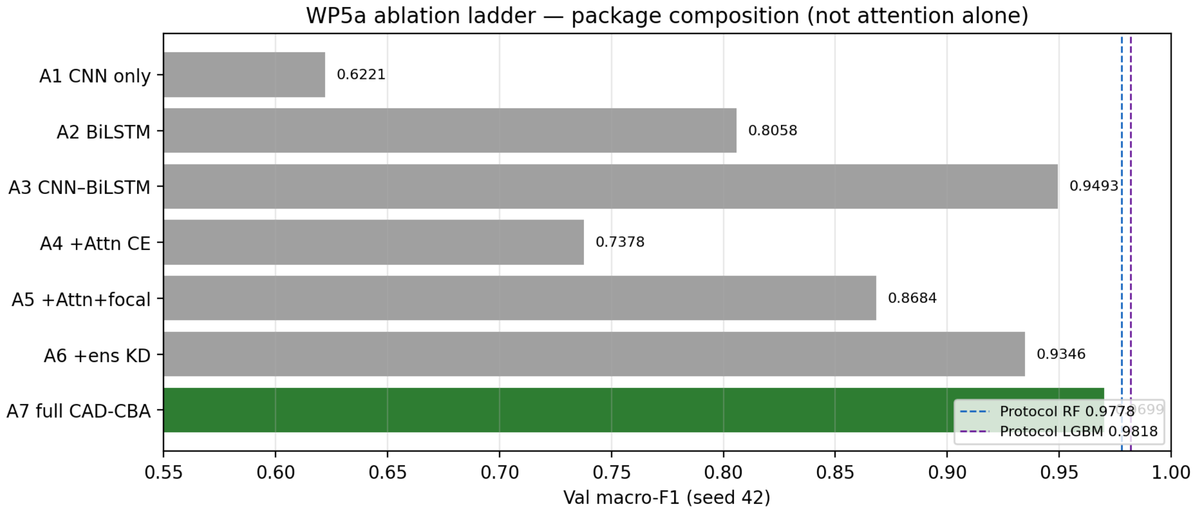

Figure 5. Ablation ladder: A7 package wins; A4 attn+CE underperforms A3.

5.5 Teachers / KD (Stage A)

Teacher	Student val F1	Decision
ensemble	0.9401	INCORPORATED
RF	0.9346	fallback
none	0.9326	control
XGB	0.9270	documented
LGBM	0.8829	weak
Neural G11 (E6)	0.8513	rejected

5.6 Multi-objective Pareto

A priori composite #1: **G6 MLP 0.9056** @ **4.33** µs/sample (F1 0.9285). F1-oriented package A7 reaches **0.9699** @ ~26 µs. Classical LGBM/RF may still lead pure F1 while neural models own deployable GPU paths.

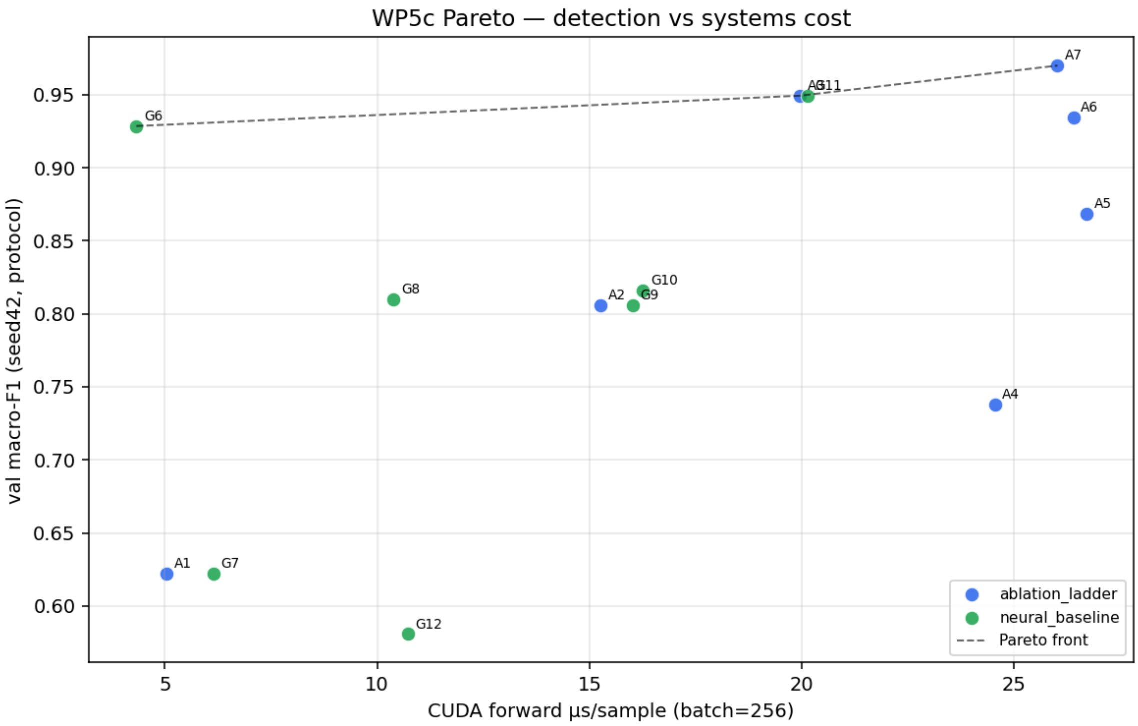

Figure 6. Pareto F1 vs latency (protocol points + classical refs).

5.7 Systems ranges (WP6b, local)

Metric	Range (session means)	Mean
Energy mJ/flow @bs=128	0.920–0.943	0.933
PT µs/sample @bs=256	24.15–25.68	24.90
CUDA derived pipeline µs	565.2–570.3	567.4
CUDA block3 FP16 µs	503.2–508.5	505.6
Peak alloc MiB	322.2	—

Table 5. Source: wp6b_local_ranges/. Do not mix with historical single-shot energy 0.786. CUDA pipeline is Option A block sum, not full V3 parity. Platform: RTX 3050 laptop only.

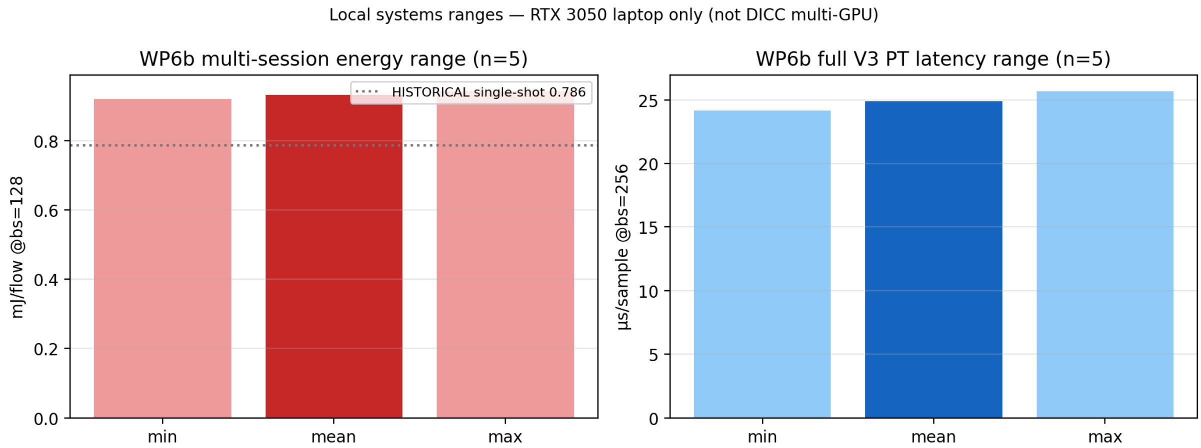

Figure 7. Local multi-session energy and PT ranges (n=5). Not DICC multi-GPU.

5.8 Fidelity (WP6a)

Export path bit-identical (max abs error 0). CUDA self-checks all PASS. Establishes operation-matched kernel correctness under Option A — not full-pipeline parity claims.

5.9 Explainability (scoped)

Metric	Value	Scope
Dispatch p99	16.60 µs	Dispatch only
Rank Spearman	0.9636	Occlusion consistency
Faith top-3 mass	0.5109	Proxy only

Metric	Value	Scope
Structured usefulness	1.0	Auto rubric n=8
Free-form feature mention	0.333	Weak → drop full claim
Gen mean	~7400 ms	Never conflate with dispatch

Table 6. J10: DROP_FULL_EXPLAINABLE_CLAIM_KEEP_STRUCTURED.

5.10 ToN honesty (13-feat)

Model	val	test
CAD-CBA-v1 (KD selected)	0.8080	0.8110
RF same-split	0.9400	0.9393

Table 7. ≠ historical clean 26-feat CNN 0.9526. Recipe transfer, not weight transfer.

5.11 Cross-GPU multi-day

V100S/A100 multi-day mean/median/std/CV/CI cells: **TBD (DICC)**. Local WP6b must not be generalised to cluster portability.

6. Discussion

Sealed multi-seed test 0.9780±0.0033 with Theft 1.0 is strong and protocol-honest, yet below protocol LGBM pure F1. That forces the lead narrative onto multi-objective validity and systems honesty — matching the Prof contribution bar. Package novelty is composition under a frozen protocol, not a single unmeasured layer. Always label val vs test, protocol vs published RF, and WP6b energy ranges vs historical 0.786. Negative results remain first-class evidence.

7. Threats to validity

1. Val vs test: most HPO/ablation numbers are val-only; sealed test is separate.
2. Dual accuracy bars: protocol RF/LGBM ≠ published RF 0.9864.
3. Single-seed ladders A1–A7 / G6–G12 vs multi-seed B14/multiruns.
4. Local ≠ portable: WP6b is RTX 3050 laptop only.
5. Option A: per-block CUDA ≠ full Custom CUDA vs full V3.
6. Energy: WP6b multi-session primary; 0.786 single-shot historical.
7. XAI: no human SOC study; free-form LLM weak; structured auto-rubric.
8. ToN: 13-feat processed recipe transfer; large RF gap.
9. Theft test support n=14 — perfect Theft F1 must state limited support.

8. Reproducibility

```
Protocol botiot_v1 · Method CAD-CBA-v1 · HPs config/hpo_best.yaml · Champion model/best_model_botiot_twostage.pth
md5 80a90f7cc210276300eaa90173a5a385 · sealed_test/ · wp6b_local_ranges/ · CLAIMS_REGISTRY.md · verify:
PYTHONPATH=. python3 scripts/verify_claims.py · Option A only · DICC not included
```

9. Conclusion

We presented CAD-CBA-v1 under a sealed, protocol-fair regime. On BoT-IoT multi-seed test, the method reaches **0.9780±0.0033** macro-F1 with Theft **1.0**, near protocol-fair RF, while LightGBM retains the pure-F1 ceiling (**0.9818** val). Ablations credit package composition. Local multi-session systems ranges quantify energy **0.920–0.943** mJ/flow, PT@256 **24.15–25.68** µs, and peak VRAM **322.2** MiB under Option A. Explainability is scoped to dispatch and structured evidence. Multi-GPU portability remains future DICC work. The primary claim is a **valid multi-objective accuracy–efficiency evaluation package** with honest limits.

Draft built 2026-07-22 from WP9b spine + on-disk JSON only. Full markdown: docs/manuscript/CAD_CBA_v1_MANUSCRIPT.md. No invented DICC numbers.